

HW 0505

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24.

(a)

$$\ln(PRICE) = \pi_0 + \pi_1 MON + \pi_2 TUE + \pi_3 WED + \pi_4 THU + \pi_5 STORMY + \pi_6 MIXED + v$$

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	106	15.876				
2	104	12.115	2	3.760	16.140	0.00000

Based on the regression outputs, the p value of variable MIXED is $0.00321 < 0.05$, making it statistically significant at the 5% level. Furthermore, the joint F -test for the exclusion of STORMY and MIXED yields an F -statistic of $16.140 > 10$ with a p -value of 0.00000. Conclude that these instruments are jointly significant and strong enough to avoid weak instrument problems.

(b)

Estimating the demand equation with both STORMY and MIXED as instruments yields a price elasticity of -0.930 (SE: 0.353), which is statistically significant at the 1% level. Compared to the exactly identified model in Table 11.5 (where the elasticity was -1.119 and SE was 0.4286), the demand becomes slightly less price-sensitive, but its standard error noticeably decreases. This drop in the standard error confirms a clear gain in asymptotic efficiency (higher precision) by introducing the second valid instrument, while the coefficients for the daily indicators (tue and wed remaining significantly negative) stay highly consistent with the original findings.

(c)

Since we now have two instruments (STORMY, MIXED) for one endogenous regressor ($\ln(PRICE)$), the model is overidentified ($L - B = 2 - 1 = 1$ degree of freedom), allowing us to test the joint exogeneity of our instruments using the Sargan test. Sargan Test Statistic (J): 0.772, p -value: 0.380. Conclusion: With a p -value of 0.380, we fail to reject the null hypothesis at any conventional significance level (such as 5%). The null hypothesis states that all instruments are uncorrelated with the demand error term (e_{dt}) and are correctly excluded from the demand equation. Therefore, the Sargan test supports the validity and exogeneity of using both STORMY and MIXED as instruments.

(d)

F -statistic: 0.6279, p -value: 0.6437 With a p -value of 0.6437, we fail to reject the null hypothesis at the 5% level. The null hypothesis states that the coefficients for the daily indicators (MON, TUE, WED, THU) are jointly equal to zero in the $\ln(PRICE)$ reduced-form equation. This implies that day-to-day shifts in demand do not have a statistically significant joint impact on the market price of fish once weather conditions are controlled for. And No, we CANNOT estimate the supply equation using 2SLS

28.

(a)

$$Q = \alpha_1 + \alpha_2 P + \alpha_3 PS + \alpha_4 DI + e_d$$

$$Q = \beta_1 + \beta_2 P + \beta_3 PF + e_s$$

Inverse Demand:

$$P = \frac{1}{\alpha_2} Q - \frac{\alpha_1}{\alpha_2} - \frac{\alpha_3}{\alpha_2} PS - \frac{\alpha_4}{\alpha_2} DI - \frac{1}{\alpha_2} e_d$$

$$P = \delta_1 + \delta_2 Q + \delta_3 PS + \delta_4 DI + u_d$$

δ_2 (coefficient of quantity): Expected to be negative ($\delta_2 < 0$). According to the law of demand, price and quantity demanded move inversely ($\alpha_2 < 0$), therefore $\delta_2 = 1/\alpha_2 < 0$.

δ_3 (coefficient of substitute price PS): Expected to be positive ($\delta_3 > 0$). An increase in the price of substitutes will increase the demand for truffles, pushing up the market price of truffles with a constant supply ($\alpha_3 > 0$, therefore $\delta_3 = -\alpha_3/\alpha_2 > 0$).

δ_4 (coefficient of disposable income DI): Expected to be positive ($\delta_4 > 0$). Truffles are generally considered a normal good; increased income will lead to increased demand, thus pushing up the market price ($\alpha_4 > 0$, therefore $\delta_4 = -\alpha_4/\alpha_2 > 0$).

Inverse Supply:

$$P = \frac{1}{\beta_2} Q - \frac{\beta_1}{\beta_2} - \frac{\beta_3}{\beta_2} PF - \frac{1}{\beta_2} e_s$$

$$P = \phi_1 + \phi_2 Q + \phi_3 PF + u_s$$

ϕ_2 (the quantity coefficient): expected to be positive ($\phi_2 > 0$). According to the law of supply, price and quantity supplied move in opposite directions ($\beta_2 > 0$), therefore $\phi_2 = 1/\beta_2 > 0$.

ϕ_3 (the coefficient of factor prices PF): expected to be positive ($\phi_3 > 0$). Increased production costs (such as factor prices) lead to a decrease in supply, which in turn pushes up the market equilibrium price with demand remaining constant ($\beta_3 < 0$, therefore $\phi_3 = -\beta_3/\beta_2 > 0$).

(b)

For the inverse demand equation, the two-stage least squares estimation yields the relationship $p = -11.428 - 2.671q + 3.461ps + 13.390di$, where all estimated signs strictly

Table 1: tsls_{demand}

	<i>Dependent variable:</i>
	p
q	-2.671** (1.175)
ps	3.461*** (1.116)
di	13.390*** (2.747)
Constant	-11.428 (13.592)
Observations	30
R ²	0.557
Adjusted R ²	0.506
Residual Std. Error	13.166 (df = 26)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

conform to economic theory. The negative coefficient on quantity matches the law of demand, while the positive coefficients on the price of substitutes and disposable income indicate that truffles are a normal good and that the alternative goods behave as expected. In terms of statistical significance, all explanatory variables are significantly different from zero, as the quantity coefficient is significant at the 5% level with a t -statistic of approximately -2.27 , while both the price of substitutes and disposable income are highly significant at the 1% level, exhibiting robust t -statistics of 3.10 and 4.87 respectively.

For the inverse supply equation, the 2SLS estimation results produce the relationship $p = -58.798 + 2.937q + 1.155pf$, where the parameter signs are completely consistent with standard supply theory. The positive slope for quantity confirms that the inverse supply curve is upward-sloping, and the positive coefficient on the factor input price accurately reflects that higher production costs reduce market supply and drive up prices. Regarding statistical significance, the coefficients are exceptionally reliable and significantly different from zero at the 1% level, with the t -statistic for quantity reaching approximately 6.22 and the t -statistic for the input price standing at around 4.70, proving that both variables exert a powerful and distinct impact within the supply structure.

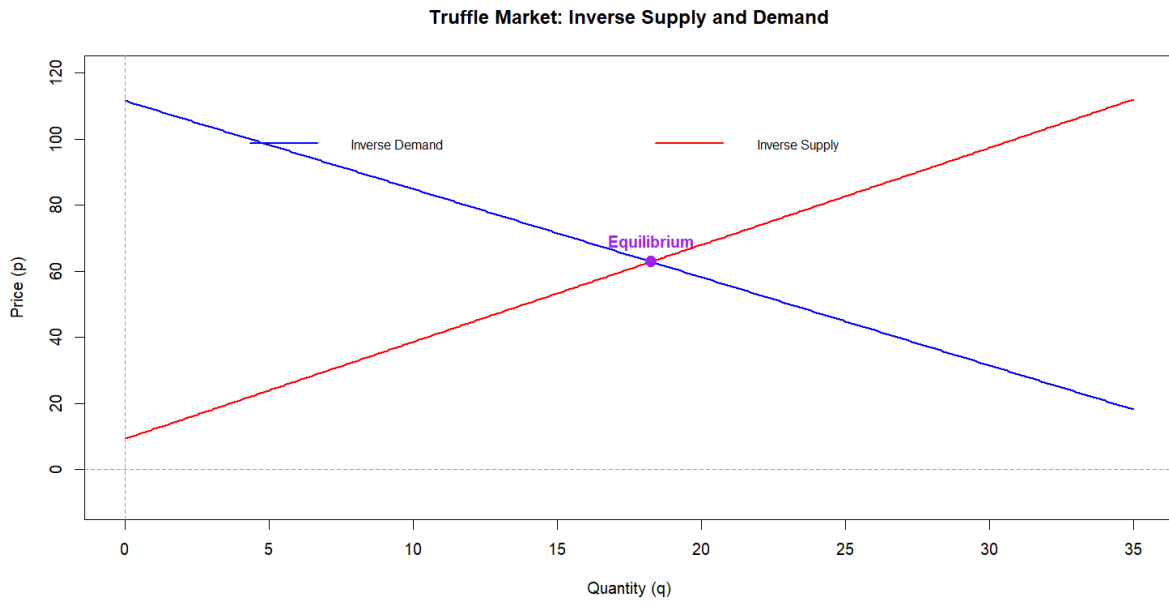
(c)

Price Elasticity of Demand at the means: -1.2725.

(d)

Table 2: tsls demand

<i>Dependent variable:</i>	
	p
q	-2.671** (1.175)
ps	3.461*** (1.116)
di	13.390*** (2.747)
Constant	-11.428 (13.592)
Observations	30
R ²	0.557
Adjusted R ²	0.506
Residual Std. Error	13.166 (df = 26)
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01	



(e)

The equilibrium values obtained from the structural equations in part (d) are $q^* = 18.2502$ and $p^* = 62.8426$. When recalculating the predicted equilibrium values using the corrected reduced-form equations from Table 11.2 with the same exogenous

values ($di^* = 3.5$, $pf^* = 23$, and $ps^* = 22$), the results are $q_{\text{reduced}} = 18.2601$ and $p_{\text{reduced}} = 62.8143$.

Comparing these new results shows that they agree exceptionally well, with only negligible differences in the second decimal place (18.25 versus 18.26 for quantity, and 62.84 versus 62.81 for price). This tight alignment is highly consistent with econometric theory because the reduced-form equations are mathematically derived directly from the structural system, meaning they should theoretically yield identical market-clearing equilibrium values for any given set of exogenous variables. The tiny remaining discrepancies are purely due to minor rounding errors in the coefficients printed in the textbook's table, confirming that algebraic restructuring and the 2SLS estimation process are entirely correct.

(f)

Table 3: ols demand	
	<i>Dependent variable:</i>
	p
q	0.151 (0.499)
ps	1.361** (0.594)
di	12.358*** (1.825)
Constant	-13.619 (9.087)
Observations	30
R ²	0.801
Adjusted R ²	0.778
Residual Std. Error	8.814 (df = 26)
F Statistic	34.952*** (df = 3; 26)
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01	

Based on the Ordinary Least Squares (OLS) estimation results provided in the tables, the estimated inverse demand equation is $p = -13.619 + 0.151q + 1.361ps + 12.358di$ and the estimated inverse supply equation is $p = -52.876 + 2.661q + 2.922pf$.

For the inverse demand equation, the estimated signs are not entirely correct according to economic theory. While the positive coefficients on the price of substitutes (1.361) and disposable income (12.358) match theoretical expectations for a normal good, the coefficient on quantity (0.151) is positive, which completely violates the law of demand. In terms of statistical significance, this incorrect quantity coefficient is not significantly

Table 4: ols supply

	<i>Dependent variable:</i>
	p
q	2.661*** (0.171)
pf	2.922*** (0.148)
Constant	-52.876*** (5.024)
Observations	30
R ²	0.953
Adjusted R ²	0.950
Residual Std. Error	4.202 (df = 27)
F Statistic	274.351*** (df = 2; 27)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

different from zero, yielding a very low t -statistic of approximately 0.30 (0.151/0.499). Conversely, the coefficients for the price of substitutes and disposable income are statistically significant at the 5% and 1% levels, respectively.

For the inverse supply equation, the estimated signs are completely correct, as the positive coefficient on quantity (2.661) confirms an upward-sloping supply curve, and the positive coefficient on the price of factor inputs (2.922) accurately reflects that higher input costs push market prices up. Both of these supply coefficients are exceptionally significant, easily passing the 1% significance threshold with robust t -statistics of 15.56 for quantity (2.661/0.171) and 19.74 for the input price (2.922/0.148).

Comparing these OLS results to the 2SLS estimates from part (b) reveals a severe simultaneous equations bias in the demand side of the OLS model. Because price and quantity are jointly determined in the truffle market, the OLS estimator fails to separate shifts in demand from shifts in supply, causing the estimated demand slope to severely bias upward into an statistically insignificant positive number (0.151). The 2SLS method successfully corrects this endogeneity bias by using valid instruments, restoring the demand slope to its theoretically correct and statistically significant negative value of -2.671. On the supply side, however, the OLS and 2SLS estimates remain relatively close and stable. This occurs because the demand equation contains two highly effective exogenous shifters, disposable income and the price of substitutes, which frequently and heavily shift the demand curve over time, tracing out a clear path for the supply curve and minimizing the impact of simultaneity bias on the supply side even under OLS.